Frequent Itemsets: Exercises

Exercise

Let T be a dataset of transaction. For an itemset $X = \{x_1, x_2, \ldots, x_k\}$, define the measure:

$$\zeta(X) = \min\{\mathrm{Conf}_T(\{x_i\} \to X - \{x_i\}) \ : \ 1 \leq i \leq k\}.$$

Say whether ζ is *monotone*, *anti-monotone* or neither one. Justify your answer.

Recall: $\mathrm{Conf}_T(\{x_i\} \to X - \{x_i\}) = \mathrm{Supp}_T(X)/\mathrm{Supp}_T(\{x_i\})$

$\zeta(\cdot)$ is monotone if for each pair of itemsets X, X'

$X \subseteq X' \Rightarrow \zeta(X) \leq \zeta(X')$

$\zeta(\cdot)$ is anti-monotone if for each pair of itemsets X, X'

$X \subseteq X' \Rightarrow \zeta(X) \geq \zeta(X')$

Consider an arbitrary $X = \{x_1, x_2, \dots, x_k\}$

and suppose (WLOG)

$$\mathfrak{Z}(X) = \mathrm{Conf}_T(\{x_i\} \longrightarrow X - \{x_i\}) \quad \text{for some } i$$

$$= \frac{\mathrm{Supp}_T(X)}{\mathrm{Supp}_T(\{x_i\})}$$

Let $X' \supseteq X$

$$\zeta(X) = \frac{\mathrm{Supp}_T(X)}{\mathrm{Supp}_T(\{x_i\})}$$

$$\geqslant \frac{\mathrm{Supp}_T(X')}{\mathrm{Supp}_T(\{x_i\})}$$ by anti-monotonicity of support

$$= \mathrm{Conf}_T(\{x_i\} \longrightarrow X' - \{x_i\})$$

$$\geqslant \min\{\mathrm{Conf}_T(\{y\} \longrightarrow X' - \{y\}) : y \in X'\}$$

$$= \zeta(X')$$ by definition of ζ

$$\Rightarrow \zeta(X) \geqslant \zeta(X')$$

$\Rightarrow \zeta(\cdot)$ is ANTI-MONOTONE

Exercise

Let T be a dataset of N *strings* over an alphabet Σ. Define the support of a string X with respect to T as: $X = X[0]X[1]X[2]\ldots\ldots$

$$\mathrm{Supp}_T(X) = \frac{|\{t \in T \ : \ X \text{ is a substring of } t\}|}{N},$$

that is, the fraction of strings of T that contain X as a substring. (We say that *X is a substring of t if it occurs in contiguous positions of t.*)

1. Identify a suitable anti-monotonicity property for $\mathrm{Supp}_T(\cdot)$.
2. Given a support threshold minsup $\in (0,1)$ let F_k be the set of strings of length k of support $\geq$ minsup. Define

$$C_{k+1} = \{Xa\,;\ X \in F_k, a \in \Sigma, \text{ and } X[1]\cdots X[k-1]a \in F_k\}.$$

Show that $F_{k+1} \subseteq C_{k+1}$.

$F_\ell \equiv$ Set of "frequent" strings of length ℓ $\forall \ell \geq 1$

Example

```
TCATTTAGAGGAAGTAAAAGTCGTAACAAGGT
GAACTGTCAAAACTTTTAACAACGGATCTCTT
TGTTGCTTCGGCGGCGCCCGCAAGGGTGCCCG
GGCCTGCCGTGGCAGATCCCCAACGCCGGGCC
TCTCTTGGCTCCAGCATCGATGAAGAACGCAG
CAGCATCGATGAAGAACGCAGCGAAACGCGAT
CGATACTTCTGAGTGTTCTTAGCGAACTGTCA
CGGATCTCTTGGCTCCAGCATCGATGAAGAAC
ACAACGGATCTCTTGGCTCCAGCATCGATGAA
CGGATCTCTTGGCTCCAGCATCGATGAAGAAC
GATGAAGAACGCAGCGAAACGCGATATGTAAT
```

Dataset T

- 11 strings
- Alphabet = {A, C, T G}

X = GAAG → Supp(X)=6/11

Obs: we don't count repetitions within the same string

$x' = GGAAGTA$ $\qquad$ $Supp_T(x') = 1/11$

① Anti-monotonicity property for $Supp_T(X)$

Strings X, X' with X SUBSTRING OF X'

$\Rightarrow Supp_T(X) \geq Supp_T(X')$

e.g. $X = GAAG$ $\quad X' = GGAAGTA$

② $C_{k+1} = \{Xa : X \in F_k, a \in \Sigma, X[1]X[2]..X[k-1]a \in F_k\}$

We are asked to show that $F_{k+1} \subseteq C_{k+1}$

Idea: adapt the argument made for

A-Priori

Consider an arbitrary $z = z[0]z[1]\cdots z[k] \in F_{k+1}$

I want to show that $z \in C_{k+1}$

Define

* $a \equiv z[k]$
* $x \equiv z[0]z[1]\ldots z[k-1]$
* $y \equiv z[1]\ldots z[k-1]\overset{=a}{z[k]}$

Eg. $z = GAAGT$

$a = T \qquad x = GAAG \qquad y = AAGT$

By anti-monotonicity (point 1)

$Z \in F_{k+1} \Rightarrow X, Y \in F_k$

$\Rightarrow Z = Xa \in C_{k+1}$ by definition of C_{k+1}

Exercise

Let T be a dataset of transactions over I. Let $X, Y \subseteq I$ be two closed itemsets and define $Z = X \cap Y$.

1. Find a relation among T_X, T_Y and T_Z (i.e., the sets of transactions containing X, Y, and Z, respectively). Justify your answer.
2. Show that Z is also closed.

① $Z \subseteq X \quad Z \subseteq Y \Rightarrow$

* Every $t \in T_X$ contains $Z \Rightarrow t \in T_Z$
* Every $t \in T_Y$ contains $Z \Rightarrow t \in T_Z$

$$T_X, T_Y \subseteq T_Z \Rightarrow T_X \cup T_Y \subseteq T_Z$$

(2) Let's show that $Z = X \cap Y$ is CLOSED

By contradiction, suppose Z is NOT CLOSED

hence $\exists$ an item a such that $a \notin Z$

$$\mathrm{Supp}_T(Z \cup \{a\}) = \mathrm{Supp}_T(Z)$$

$\Rightarrow$ a occurs in EVERY $t \in T_Z$

$\Rightarrow$ (by point (1)) a occurs in EVERY $t \in T_X$ and a occurs in EVERY $t \in T_Y$

$\Rightarrow$ a must belong to X since the fact that X is closed implies that there cannot be an item not in X which occurs in every transaction when X occurs

Similarly: a must belong to Y

$\Rightarrow a \in X \cap Y = Z$

which is a contradiction since we assumed $a \notin Z$

Exercise

Let T be a dataset of N transactions over a set of items I, and let $\epsilon > 0$ be a parameter. For each itemset X, let s_X be an approximation of its true support $\mathrm{Supp}_T(X)$ such that

$$\mathrm{Supp}_T(X) - \epsilon \le s_X \le \mathrm{Supp}_T(X) + \epsilon$$

Consider an ordering of all itemsets $X_1, X_2, X_3, \ldots$ by non-increasing approximate support ($s_{X_1} \ge s_{X_2} \ge s_{X_3} \ldots$), and let $K < K'$ be two indices such that $s_{X_K} > s_{X_{K'}} + 2\epsilon$.

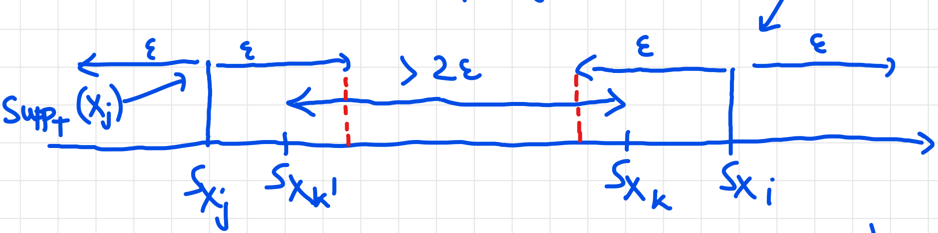
1. Show that if $i \le K$ and $j \ge K'$, then $\mathrm{Supp}_T(X_i) > \mathrm{Supp}_T(X_j)$
2. Based on the previous points, show that the set $\{X_1, X_2, \ldots, X_{K'}\}$ contains the Top-K frequent itemsets with respect to the true support.

① $X_1 \ X_2 \ X_3 \cdots X_i \cdots X_k \cdots X_{k'} \cdots X_j \cdots$

$S_{X_1} \geq S_{X_2} \geq S_{X_3} \geq S_{X_i} \geq S_{X_k} \qquad S_{X_{k'}} \qquad S_{X_j}$

$S_{X_k} > S_{X_{k'}} + 2\varepsilon$

$\mathrm{Supp}_T(X_i) > \mathrm{Supp}_T(X_j)$



$\mathrm{Supp}_T(X_i) \geq S_{X_i} - \varepsilon > S_{X_j} + \varepsilon \geq \mathrm{Supp}_T(X_j)$

$\Rightarrow \mathrm{Supp}_T(X_i) > \mathrm{Supp}_T(X_j)$

② We need to show that $X_1 X_2 \cdots X_{k'}$ contain the Top-K frequent itemsets w.r.t. true support

Obs. If Y is one of the top-k frequent itemsets (w.r.t. true supports) there are at most $k-1$ other itemsets with support STRICTLY GREATER than the support of Y

By Point ① for every X_j with $j \geq k'$ there are at least k itemsets with $(X_1 X_2 \cdots X_k)$

true support $> \text{Supp}_T(X_j)$

$\Rightarrow$ for sure X_j cannot belong to the Top-K frequent itemsets.

$\Rightarrow$ The Top-K frequent itemsets must be a subset of $X_1 X_2 \cdots X_K \cdots X_{K'}$